

k_davey_mom_keltner - Strategy Walkthrough

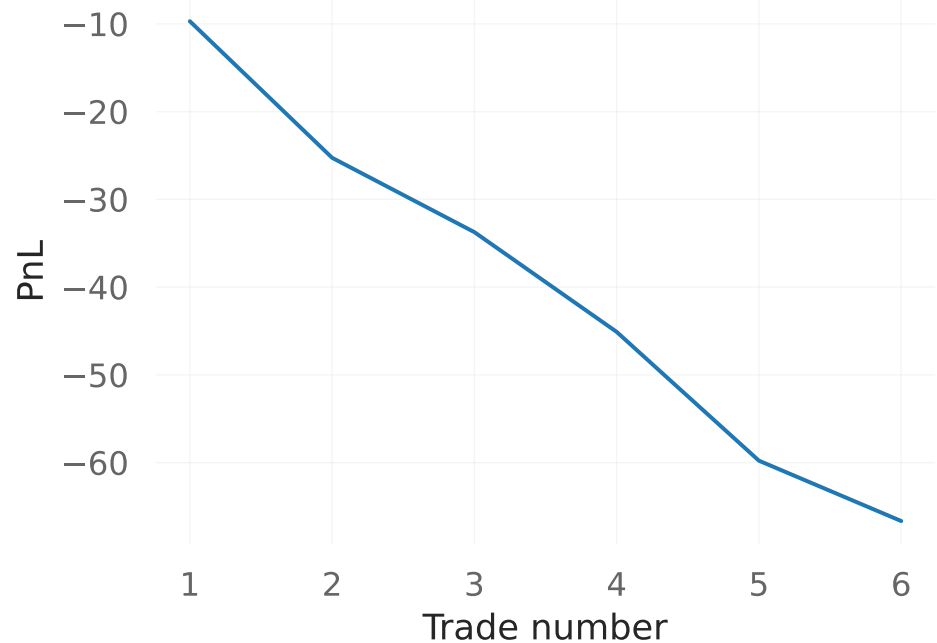
Strategy name k_davey_mom_keltner Files reviewed - src/strategies/k_davey_mom_keltner/strate
src/strategies/k_davey_mom_keltner/backtest_k_davey_mom_keltner_v2.py Step-by-step execution
Backtest fetches OHLCV and warns if timeframe is not 1d. 2) It computes keltner_stoch, trend
stops, and ATR volatility filter plus its rolling average. 3) Base equity and contract multipl
(special handling for MES symbol). 4) BacktesterV2 is initialized in futures pnl_mode with AT
multipliers by side. 5) Main loop starts after the maximum warmup across momentum, Keltner, A
and volatility windows. 6) Pending actions are executed at next bar open (ENTRY or EXIT). 7)
position exists and ATR is valid, stop is updated from average price via bt.update_stop_from
bt.on_bar checks intrabar stop execution. 9) Net equity and dynamic contract count are comput
bt.trade_size is updated each bar. 10) Entry conditions evaluate momentum, trend, volatility
stochastic thresholds. 11) Exit conditions set next-bar EXIT when Keltner stochastic exceeds
specific exit threshold. 12) Outputs are generated: stats, equity curve, optional report, and
Parameters and defaults - Momentum and Keltner: mom_length_long=40, mom_length_short=40,
keltner_length=5, keltner_atr_mult=0.5. - Thresholds and trend: entry_threshold=30.0,
exit_threshold=70.0, trend_ema=200. - Volatility filter: volatility_atr_period=14,
volatility_sma_period=100, volatility_mult=0.8. - Position sizing: use_position_sizing=True,
base_equity=15000.0, sizing_factor=0.33, max_contracts=15. - Execution and costs: initial_bal
position_mode="fixed", commission_pct=0.001, slippage_pct=0.001. - Direction and risk: allow
atr_period=14, atr_sl_mult_long=4.0, atr_sl_mult_short=1.0, pyramiding=1, position_pct=None.
Indicators/calculations - Keltner stochastic proxy: $100 * (\text{close} - \text{lower_kc}) / (\text{upper_kc} - \text{lo}$
Momentum checks: close vs close N bars ago (separate N for long/short). - trend_ema filter fr
ATR for stop updates and initial ATR-based risk. - Volatility filter: $\text{ATR}(\text{volatility_atr_peri}$
 $\text{rolling_mean_ATR} * \text{volatility_mult}$. - Dynamic contracts from net_equity/base_equity formula (
position_pct path). Entry conditions - LONG setup requires all: - price > close[mom_length
ago] - price > trend_ema - volatility filter true - keltner_stoch < entry_threshold -
position, or existing LONG with room for pyramiding. - SHORT setup (if allow_short) mirrors l
price < close[mom_length_short bars ago] - price < trend_ema - volatility filter true -
keltner_stoch > (100 - entry_threshold) - no position, or existing SHORT with room for pyra
Entries are queued and executed on next bar open. Exit conditions - If LONG and keltner_stoc
exit_threshold, queue EXIT for next bar open. - If SHORT and keltner_stoch < (100 - exit_thre
queue EXIT for next bar open. - Additional exits can happen intrabar from ATR stop handling i
BacktesterV2. Risk and position management (SL/TP/ATR/trailing/pyramiding/position sizing) -
based stop distances with different multipliers for LONG and SHORT. - No fixed percent SL/TP
runner (both set to None in BacktesterV2 init). - Dynamic position size (contracts) based on
progression, capped by max_contracts. - Optional position_pct path overrides formula-based si
Pyramiding supported and checked with len(bt.lots) < bt.pyramiding. - Futures PnL mode with c
multiplier (5 for MES, else 50 default path). Outputs (trades, stats, plots, reports) - clea
bt.stats(). - Equity curve PNG under static/backtests/k_davey_mom_keltner/<run_id>/. - Trades
extensive strategy metadata. - QuantStats report when enabled. Known ambiguities or inconsis
the code - Function signature includes generate_plots but no trade-plot generation is impleme
base_equity default in signature is 15000.0, but fallback block uses 1500.0 for MES only when
is None. - Parameters named ema_fast/ema_slow appear in CSV schema although this strategy doe
EMA crossover periods.

k_davey_mom_keltner - Trade summary

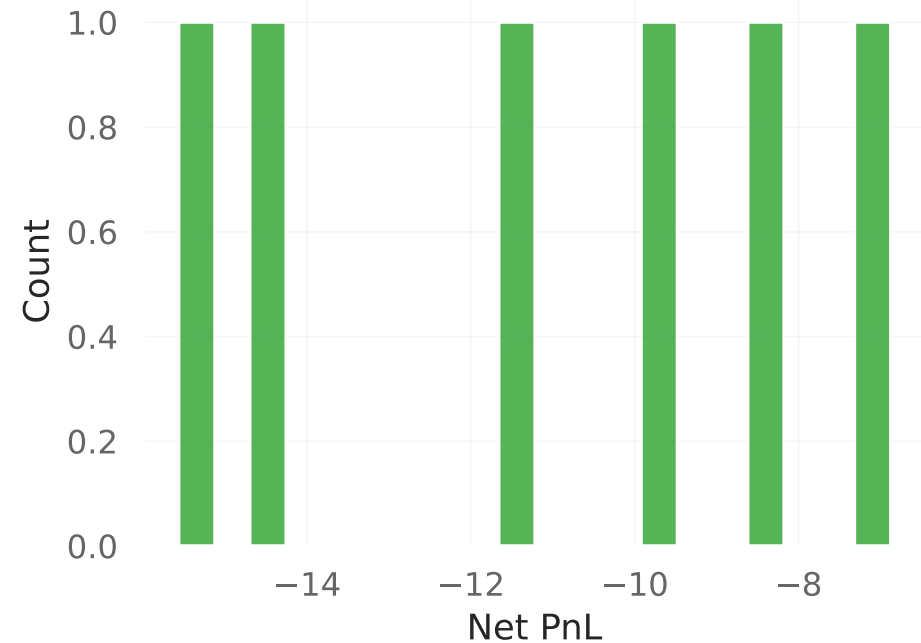
Trades: 6 Total Net PnL: -66.65 Average Net PnL per trade: -11.11 Win rate: 0.00% First trigger: Momentum+Keltner long Last trigger: Keltner stoch exit Source CSV: docs/strategy_walkthroughs/csv_baselines/k_davey_mom_keltner.csv

k_davey_mom_keltner - Trade analytics

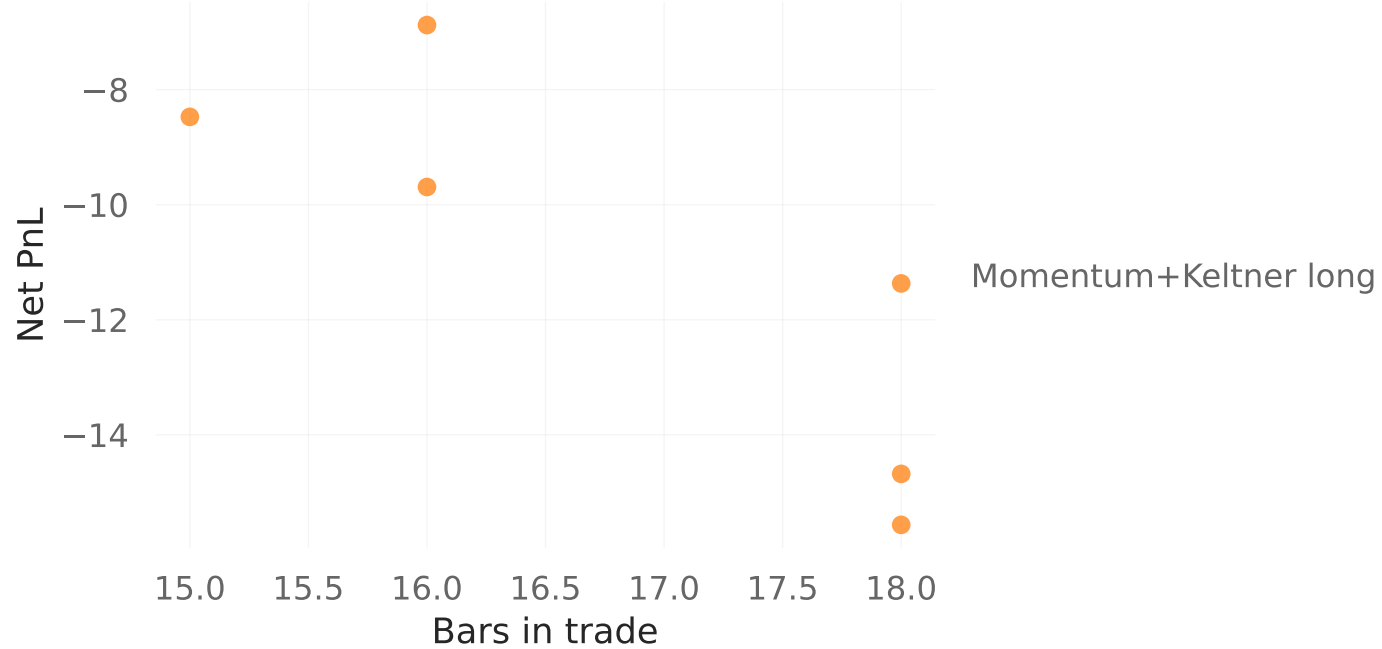
Cumulative Net PnL by trade



Net PnL distribution



Duration vs Net PnL



Top entry triggers

